Data Analysis

DATA ANALYSIS

The collection, transformation, and organization of data in order to draw conclusions, make predictions, and drive informed decision making

DATA IS EVERYWHERE

You use and create data everyday.
The reviews of a product you buy online to decide whether or not to buy it. That's data analysis.
Or maybe the fitness tracker that counts your step to count calories burned throughout the day.

That is DATA ANALYSIS.

DATA IS EVERYWHERE

But you don't just use data, you also create huge amounts of it every single day. Anytime you use your phone, look up something online, stream music, shop with a credit card, etc. you are creating data.

WHAT DO DATA ANALYST DO?

Data Analyst is someone who:

- Collects
- Transforms
- Organize

To help make informed decisions

DATA ANALYSIS PROCESS

1	2	3	4	5	6
ASK	COLLECT	CLEAN	ANALYSE	VISUALISE	ACT
Define the problem & research question	Gather data from reliable sources	Fix errors, remove duplicates, handle nulls	Apply statistics & techniques to explore	Create charts & communicate findings	Make decisions & recommendations

⚠ This is NOT a straight line — it's a loop. You will often revisit earlier steps as you learn more about your data. Real projects rarely flow neatly from Step 1 to Step 6.

Where analysts typically spend their time:

STEPS 1 & 2 – ASK THE RIGHT QUESTION · COLLECT THE RIGHT DATA

1 ASK

Define the problem before touching the data.

- What decision will this analysis support?
- Who is the audience for this analysis?
- What does a useful answer look like?
- What data would I need to answer this?
- What are the constraints? (time, data access)

🔑 Poor question → poor analysis. 'What is our data?' is not a question. 'Which products have declining repeat-purchase rates among customers under 30?' is.

2 COLLECT

Data quality determines analysis quality.

Primary Data

Surveys, interviews, experiments, sensors — data you collect yourself

Secondary Data

Published databases, government records, company archives, web scraping

Tertiary Data

Aggregated sources — research papers, meta-analyses, market reports

Ask: Is it Reliable? Accurate? Complete? Current? Relevant? (RACCR)

STEP 3 – CLEAN THE DATA (THE MOST UNDERESTIMATED STEP)

What Does 'Dirty' Data Look Like?

❌ **Missing values**
Student age: blank, NULL, N/A, 999

❌ **Duplicates**
Same customer record appears 3 times

❌ **Inconsistent format**
Date as '04/03/26', '2026-03-04', 'March 4'

❌ **Typos & errors**
'Tokoy', 'TOKYOO', 'tokyo' all meaning Tokyo

❌ **Outliers / implausible**
Age: 312, Salary: -¥50,000, Height: 0 cm

❌ **Wrong data type**
Phone number stored as decimal: 0312345678.0

How Do We Fix It?

✅ **Imputation**
Replace missing values with the mean, median, or most frequent value — or flag for removal

✅ **Deduplication**
Use unique identifiers (ID, email) to find and merge or delete duplicate records

✅ **Standardisation**
Convert all values to a single agreed format before analysis

✅ **Fuzzy matching**
Use string similarity algorithms to identify 'Tokoy' as 'Tokyo'

✅ **Outlier handling**
Investigate outliers: are they data entry errors, or genuine rare events? Remove or cap as appropriate

✅ **Type conversion**
Parse and recast columns: phone numbers as strings, dates as date objects

STEP 4 – ANALYSE THE DATA (TURNING CLEAN DATA INTO ANSWERS)

Analytical Approaches

Descriptive Analysis
What happened?
Summarise data using measures of centre (mean, median, mode) and spread (range, standard deviation, variance). Build frequency tables and summary statistics.

Exploratory Analysis (EDA)
What patterns exist?
Use visualisations and statistics to freely explore the dataset — spot trends, anomalies, clusters, and relationships before testing any hypothesis.

Diagnostic Analysis
Why did it happen?
Dig deeper into patterns to identify causes. Compare subgroups, test variables, look for correlations and contributing factors.

Predictive Analysis
What is likely to happen?
Use statistical models or machine learning to forecast future outcomes based on historical patterns. (Advanced — covered later in the course.)

Key Statistical Concepts

Mean
Sum of all values ÷ number of values. Sensitive to outliers.
e.g. Scores: 70, 80, 90, 10 → Mean = 62.5 (pulled down by outlier)

Median
Middle value when sorted. Robust against outliers.
e.g. Scores: 10, 70, 80, 90 → Median = 75

Mode
Most frequently occurring value. Useful for categorical data.
e.g. Survey: 'Satisfied' appears 42 times → Mode = Satisfied

Standard Deviation
Average distance of each data point from the mean. Measures spread.
e.g. Low SD = data clustered tightly; High SD = data widely spread

Correlation (r)
Measures strength & direction of a linear relationship. Range: –1 to +1.
e.g. r = 0.9 = strong positive; r = –0.7 = strong negative; r ≈ 0 = none

STEP 5 – VISUALISE THE DATA

Bar / Column Chart

Use for: Comparing categories

When: How do sales differ across 5 regions?

⚠ *Don't use for time trends with many data points*

Line Chart

Use for: Showing trends over time

When: How did revenue change month by month?

⚠ *Don't use for unordered categories*

Scatter Plot

Use for: Exploring relationships between two variables

When: Is there a link between study hours and score?

⚠ *Don't use with fewer than ~20 data points*

Histogram

Use for: Showing distribution of a single variable

When: How are exam scores distributed across the class?

⚠ *Don't confuse with bar charts — no gaps between bars*

Pie / Donut Chart

Use for: Showing parts of a whole (use sparingly)

When: What share of revenue comes from each product?

⚠ *Never use with more than 5–6 segments; hard to compare*

Heatmap / Table

Use for: Showing magnitude across two dimensions

When: Which product sold most in which month?

⚠ *Can be hard to read without clear colour scale*

STEP 6 – ACT ON THE DATA (WHERE ANALYSIS CREATES REAL VALUE)

✅ Make a Decision

The analysis directly answers a yes/no or which-option question. A recommendation is made and executed.

- Launch Product B, not Product A — data shows 2× higher demand
- Hire 3 additional nurses for the night shift — patient wait times exceed safe thresholds
- Stop running the Tuesday ad campaign — cost-per-acquisition is 4× the threshold

🔄 Change a Process

The analysis reveals inefficiency or failure that requires redesigning how something works.

- Redesign the checkout flow — data shows 70% of users abandon at the payment screen
- Reschedule maintenance to weekends — failure rates spike on Monday mornings
- Change stock replenishment trigger — current threshold causes stockouts 3× per month

❓ Raise More Questions

A legitimate and common outcome. Good analysis often surfaces new unknowns that require further investigation.

- Sales are up — but only in one region. Why? Launch a new data collection effort
- Churn dropped after the redesign. But is it sustained? Monitor for 6 months
- The model predicts X — but what happens if assumption Y is wrong? Run a sensitivity test

TYPES OF DATA

📐 QUANTITATIVE

Numerical — can be measured or counted

DISCRETE

Countable whole numbers. Gaps exist between values.

Examples: number of students in a class (28, 29, 30 — never 28.5)
number of cars sold, number of errors in code

Chart types: Bar chart, histogram

CONTINUOUS

Measurable values. Can take any value within a range.

Examples: height (173.4 cm), temperature (36.8 °C)
exam time (47 min 23 sec), weight, distance

Chart types: Line chart, scatter plot, histogram

🏷️ QUALITATIVE

Categorical — describes qualities or characteristics

NOMINAL

Named categories. No natural order or ranking.

Examples: eye colour, country of birth, job title
favourite music genre, blood type (A, B, O, AB)

Chart types: Pie chart, bar chart, word cloud

ORDINAL

Ordered categories. Direction is clear; gaps are not equal.

Examples: satisfaction rating (1–5 stars)
education level (high school < bachelor < master < PhD)
military rank, pain scale (0–10)

Chart types: Bar chart, Likert scale chart

TOOLS USED IN DATA ANALYSIS

Spreadsheets

Microsoft Excel	Beginner–Intermediate	Universal, powerful for small–medium datasets. PivotTables, charts, VLOOKUP, basic stats.
Google Sheets	Beginner	Cloud-based, free. Great for collaboration. Lighter than Excel for large data.

Programming Languages

Python (pandas, matplotlib)	Intermediate	Most popular for data analysis. Huge ecosystem, readable syntax, works with any data size.
R	Intermediate	Designed for statistics. Dominant in academic research and healthcare analytics.

Visualisation Tools

Tableau / Power BI	Beginner–Intermediate	Drag-and-drop dashboards. Industry standard. No coding needed for basic use.

DATA ETHICS

🔒 Privacy

People whose data you use have rights. In many countries (GDPR in Europe, APPI in Japan), collecting, storing, or sharing personal data without consent is illegal — and unethical regardless of legality.

⚖️ Bias & Fairness

If training data reflects historical inequalities (e.g. past hiring discrimination), models built on that data will perpetuate those inequalities. Analysts must actively look for and report bias.

🎯 Transparency

Your audience should understand what data you used, how you analysed it, and what assumptions you made. 'Black box' conclusions are not acceptable in high-stakes decisions.

✔️ Consent & Purpose

Data collected for one purpose (e.g. medical treatment) should not be used for another (e.g. insurance pricing) without explicit consent. Scope creep in data use is a serious ethical issue.

🔍 Reproducibility

Your analysis should be documented so that others can check, reproduce, or challenge your conclusions. Undocumented methods undermine trust in findings.

QUESTIONS?